

EE104 Fundamentals of Electric Circuits (Lesson 13)

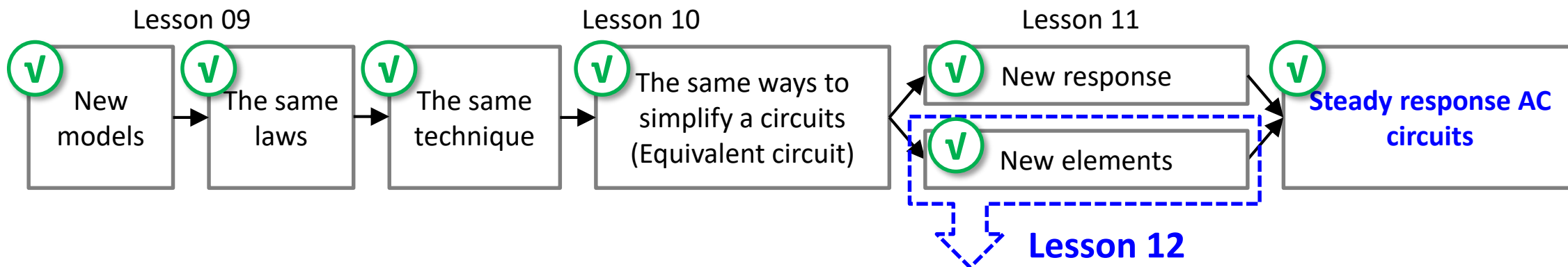
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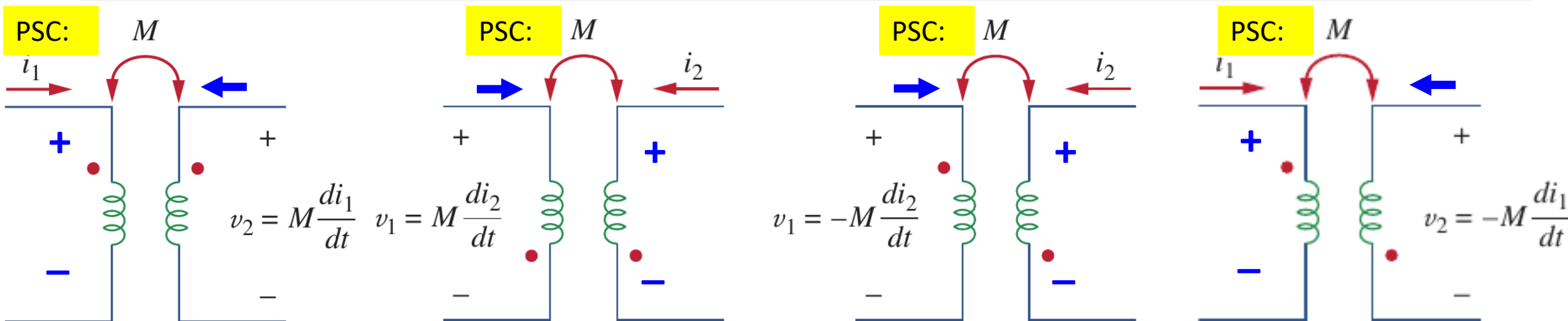
南方科技大学-工学院-电子与电气工程系-PN Lab

Review of Lessons 12

- EE104 road map

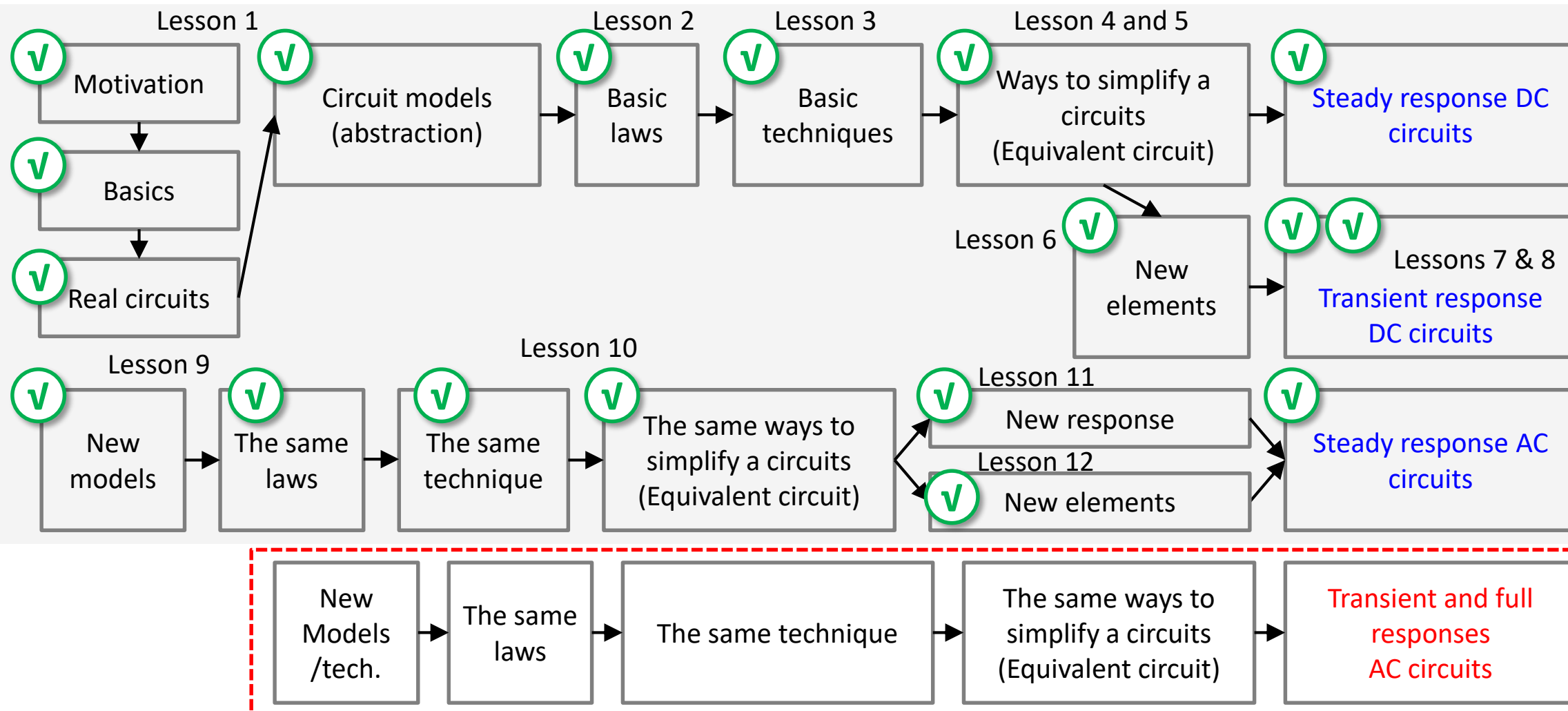


Transformers: coupling, mutual inductance, linear transformers, ideal transformers



Welcome to section 3

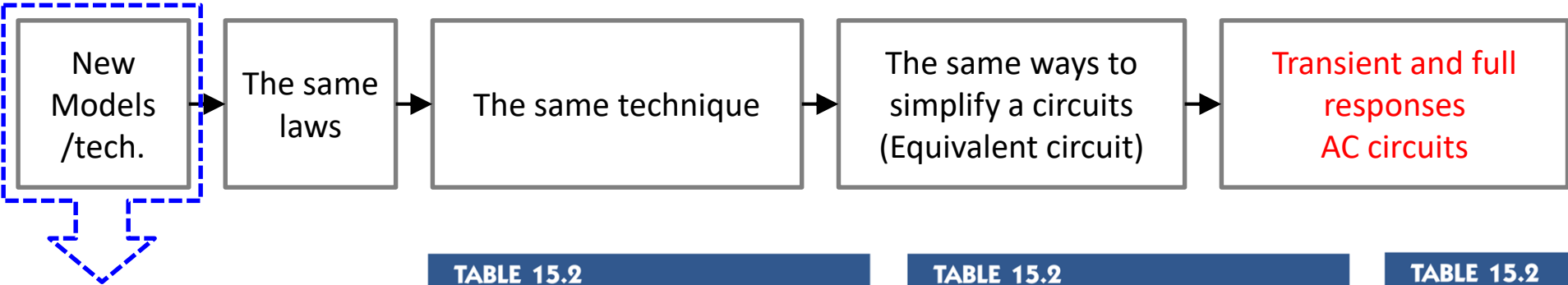
- EE104 road map



Welcome to section 3

- EE104 road map

Lesson 13



- Laplace transform
- Inverse Laplace transform

TABLE 15.2	
Laplace transform pairs.*	
$f(t)$	$F(s)$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
e^{-at}	$\frac{1}{s+a}$
t	$\frac{1}{s^2}$
t^n	$\frac{n!}{s^{n+1}}$

TABLE 15.2	
Laplace transform pairs.*	
$f(t)$	$F(s)$
te^{-at}	$\frac{1}{(s+a)^2}$
$t^n e^{-at}$	$\frac{n!}{(s+a)^{n+1}}$
$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$

TABLE 15.2	
Laplace transform pairs.*	
$f(t)$	$F(s)$
$\sin(\omega t + \theta)$	$\frac{s \sin \theta + \omega \cos \theta}{s^2 + \omega^2}$
$\cos(\omega t + \theta)$	$\frac{s \cos \theta - \omega \sin \theta}{s^2 + \omega^2}$
$e^{-at} \sin \omega t$	$\frac{\omega}{(s+a)^2 + \omega^2}$
$e^{-at} \cos \omega t$	$\frac{s+a}{(s+a)^2 + \omega^2}$

Lesson 13 Introduction to the Laplace transform

13.1-13.2	Introduction	Slides 06-13
13.3	Laplace transform: what is it?	Slides 15-26
13.4-13.5	Properties of Laplace transform	Slides 28-42
13.6	Inverse Laplace transform	Slides 44-67
13.7	*Convolution Integral	Slides 69-75
13.8	Summary	Slides 77-78

13.1 Introduction (a)

- What we have learned up to now:

Can we transform this into algebraic equations using sth like phasors?

Steady-state DC circuits

Algebraic equation
(Ohm's Law, KCL, KVL)

Steady-state response

Transients in DC circuits

First-order and second-order
differential equations

Steady-state + transient responses

Steady-state AC circuits

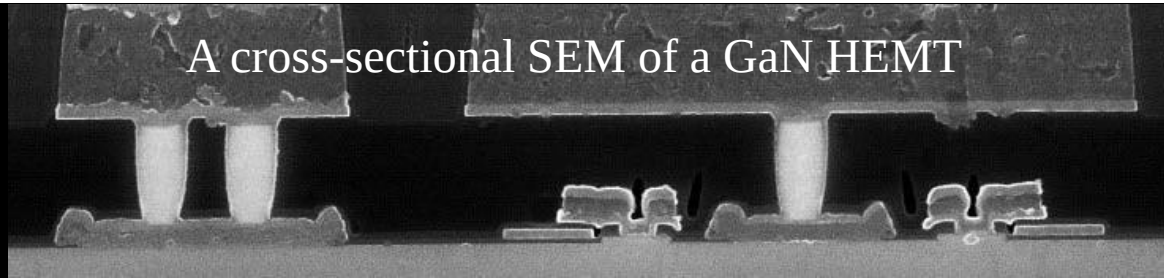
Algebraic equation due to
phasors: the big three again

Steady-state response

What is the full response of AC circuits?

13.1 Introduction (b)

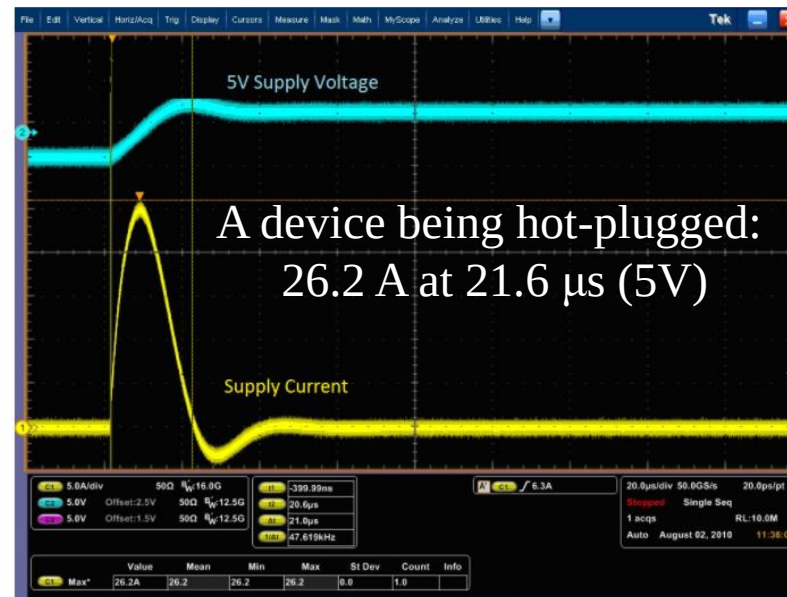
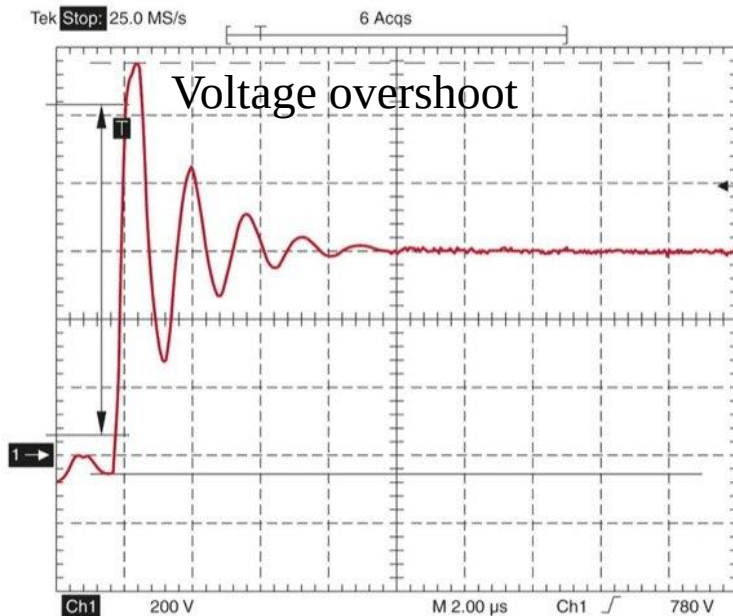
- The complete response is important: an example



- a) Fast charging is now popular.
- b) While the charger is used for 220 V, the core devices inside, i.e. power GaN HEMTs, are rated for 650 V.
- c) As a matter of fact, the breakdown voltage of such devices can be even higher than 1000 V.
- d) Why do we need such high-voltage devices for a voltage of only 220 V

13.1 Introduction (c)

- The complete response is important: an example
 - a) For sake of safety, engineers need to know the response of a circuit to ensure the current/voltage rating of components are not exceeded at any instant of time.
 - b) This means one needs to know the complete response.



For example: a voltage transient can be anywhere from a few millivolts to thousands of volts, and they can last from nanoseconds to hundreds of milliseconds.

Ways to suppress: bypass capacitors, Zener diodes, TVS diodes, MOVs, etc.

13.1 Introduction (d)

- Laplace transform for circuit analysis:

- a) In this lesson, we are going to learn a generic approach applicable to wide variety of inputs beyond the aforementioned three types.
- b) Similar to the application of phasor transform to solve the steady state AC circuits, Laplace transform can be used to transform the time domain circuits into s domain circuits to simplify the solution of integral differential equations to the manipulation of a set of algebraic equations.

(1) Phasors:

From time domain to frequency domain

$$u(t) = Ri + L \frac{di}{dt} + \frac{1}{C} \int i dt$$

$$\Rightarrow \mathbf{V} = R\mathbf{I} + j\omega L\mathbf{I} + \frac{\mathbf{I}}{j\omega L}$$

(2) Laplace transform:

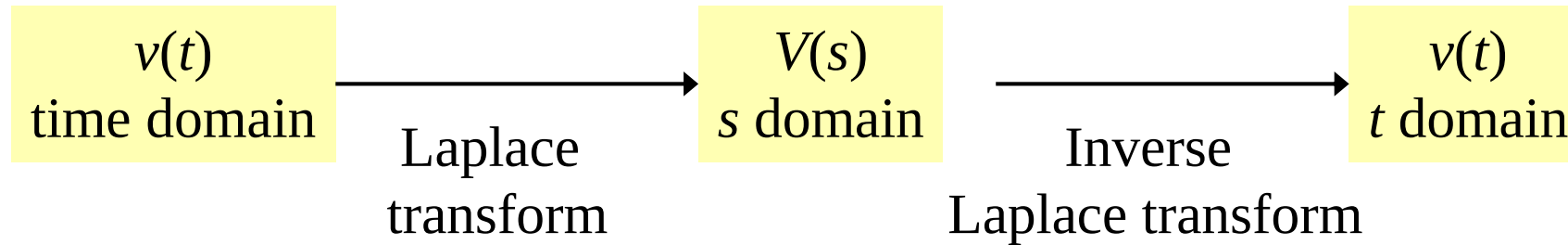
From time domain to s domain

$$u(t) = Ri + L \frac{di}{dt} + \frac{1}{C} \int i dt$$

$$\Rightarrow V(s) = RI + sLI + \frac{I}{sC}$$

13.1 Introduction (e)

- Laplace transform for circuit analysis:



The Laplace transform is of vast importance:

1. It can be applied to a wider variety of inputs (non-sinusoidal) than phasor analysis.
2. It provides an easy way to solve circuit problems involving initial conditions.
3. The total response can be obtained in a single step.
4. Transfer function defined in s-domain reflects the circuit properties such as stability.

13.2 Interesting to know (a)

13.2 Interesting to know (b)

13.2 Interesting to know (c)

- Integration by parts (分部积分法)

$$\begin{aligned}\int_a^b u(x)v'(x) dx &= [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) dx \\ &= u(b)v(b) - u(a)v(a) - \int_a^b u'(x)v(x) dx\end{aligned}$$

Proof: $\frac{d}{dx} (u(x)v(x)) = v(x) \frac{d}{dx} (u(x)) + u(x) \frac{d}{dx} (v(x)).$

Integrating both sides with respect to x ,

$$\int \frac{d}{dx} (u(x)v(x)) dx = \int u'(x)v(x) dx + \int u(x)v'(x) dx$$

then applying the definition of [indefinite integral](#),

$$u(x)v(x) = \int u'(x)v(x) dx + \int u(x)v'(x) dx$$

Lesson 13 Introduction to the Laplace transform

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13.3 Laplace transform (a)

- Laplace transform:

Definition: Given a function $f(t)$, $t \geq 0$, its Laplace transform $F(s) = \mathcal{L}\{f(t)\}$ is defined as

$$F(s) = \mathcal{L}\{f(t)\} \doteq \int_0^{\infty} e^{-st} f(t) dt \doteq \lim_{A \rightarrow \infty} \int_0^A e^{-st} f(t) dt$$

We say the transform converges if the limit exists, and diverges if not.

- a) The Laplace transform is an integral transformation of a function $f(t)$ from the time domain into the complex frequency domain, giving $F(s)$.
- b) s is a complex variable given by $s = \sigma + j\omega$ and has the dimensions of frequency and unit of Hertz.
- c) Note that the integral only exists for certain range of s (range of convergence).
- d) The lower limit is 0^- to indicate a time just before $t = 0$. We use 0^- this to include the origin and capture any discontinuity of $f(t)$ at $t = 0$, accommodating functions that may be discontinuous at $t = 0$ (such as singularity functions).

13.3 Laplace transform (b)

- Unilateral and bilateral Laplace transform:

- a) The unilateral Laplace transform is adequate for this course and is the only Laplace transform that we discuss and treat in the following slides.
- b) $f(t)$ is sometimes called 原函数 and $F(s)$ is sometimes called 象函数.

One-sided Laplace transform
(Unilateral Laplace transform)

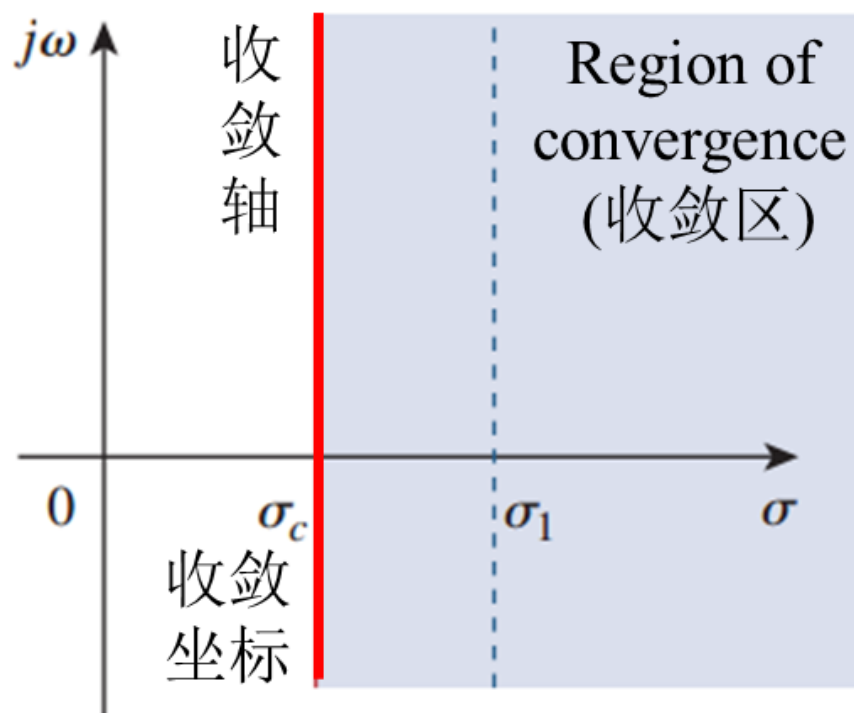
$$\mathcal{L}[f(t)] = F(s) = \int_{0^-}^{\infty} f(t)e^{-st} dt$$

Two-sided Laplace transform
(Bilateral Laplace transform)

$$F(s) = \int_{-\infty}^{\infty} f(t)e^{-st} dt$$

13.3 Laplace transform (c)

- Convergence (收敛):



a) Region of convergence (ROC, 收敛区): a region in the s-domain in which $F(s)$ has finite values

$$\int_0^\infty |f(t)| e^{-\sigma t} dt = \int_0^\infty |f(t)| e^{-\sigma t} |e^{-j\omega t}| dt < \infty$$

b) This indicates that $\int_0^\infty |f(t)| e^{-\sigma t} dt < \infty$, since

$$|e^{-j\omega t}| = \sqrt{\cos^2 \omega t + \sin^2 \omega t} = 1$$

13.3 Laplace transform (d)

- Convergence (收敛):

1. For a function $f(t) = e^{\alpha t}$, we have

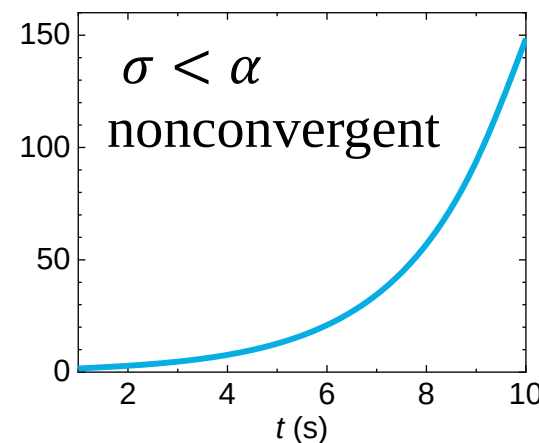
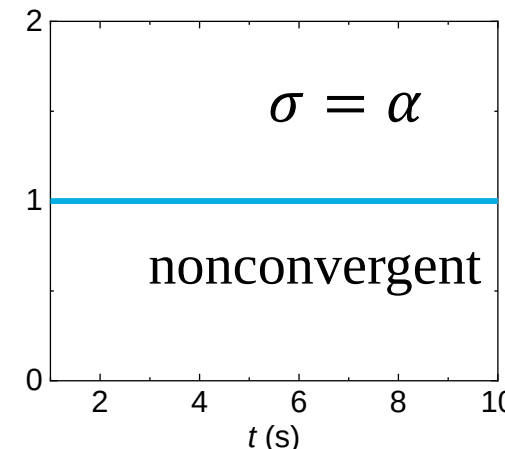
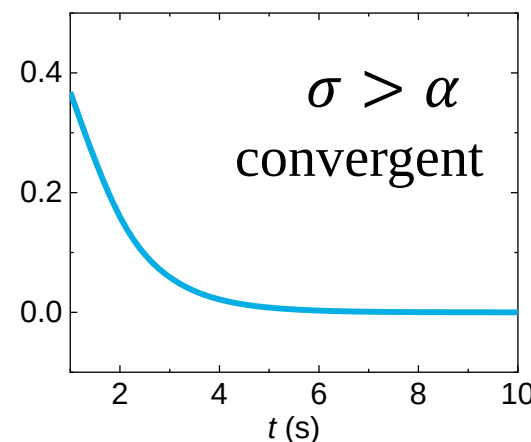
$$\int_{0^-}^{\infty} e^{\alpha - \sigma t} dt < \infty$$

therefore the ROC is $\sigma > \alpha$

2. For another function $f(t) = e^{t^2}$, we need

The function increases with t at a much greater rate than the decreasing rate of the exponential function $e^{-\sigma t}$ (Laplace kernel), as a result, the integral does not converge for any s .

$$f(t) = e^{(\alpha - \sigma)t}$$



If $f(t)$ increases at a slower rate as compared to $e^{-\sigma t}$, the transform is convergent. Otherwise it is not convergent.

13.3 Laplace transform (e)

- The most important Laplace transform in the world:

$$\text{If } f(t) = e^{at},$$

$$F(s) = \mathcal{L}(e^{at}) = \int_{0^-}^{\infty} e^{at} e^{-st} dt = \int_{0^-}^{\infty} e^{-(s-a)t} dt$$

$$= \left[-\frac{1}{s-a} e^{-(s-a)t} \right]_0^{\infty}$$

$$= \frac{1}{s-a} \quad (s > a)$$

$$f(t) = e^{at}$$



$$F(s) = \frac{1}{s-a}$$

13.3 Laplace transform (f)

- Example 13.01

Example 13.01: determine the Laplace transform of each of the following functions:

(a) $u(t)$, (b) $e^{-at}u(t)$, (c) $\delta(t)$, (d) $Ae^{-at}u(t)$ and (e) $Be^{-j\omega t}u(t)$.

Solution:

$$(a) \mathcal{L}(u(t)) = F(s) = \int_{0^-}^{\infty} 1e^{-st} dt = \left[-\frac{e^{-st}}{s} \right]_0^{\infty} = -\frac{1}{s}(0) + \frac{1}{s}(1) = \frac{1}{s}$$
$$(b) \mathcal{L}(e^{-at}u(t)) = F(s) = \int_{0^-}^{\infty} 1e^{-(s+a)t} dt = \left[-\frac{1}{s+a} e^{-(s+a)t} \right]_0^{\infty} = \frac{1}{s+a}$$
$$(c) \mathcal{L}(\delta(t)u(t)) = F(s) = \int_{0^-}^{\infty} \delta(t)e^{-st} dt = e^{-0} = 1$$

13.3 Laplace transform (g)

- Example 13.01

Example 13.01: determine the Laplace transform of each of the following functions: (a) $u(t)$, (b) $e^{-at}u(t)$, (c) $\delta(t)$, (d) $Ae^{-at}u(t)$ and (e) $Be^{-j\omega t}u(t)$.

Solution: (d) $F(s) = \mathcal{L}(Ae^{-at}u(t)) = \int_{0^-}^{\infty} Ae^{-(s+a)t} dt = \left[-A \frac{e^{-(s+a)t}}{s+a} \right]_0^{\infty} = \frac{A}{s+a}$

(e) $F(s) = \mathcal{L}(Be^{-j\omega t}u(t)) = \int_{0^-}^{\infty} Be^{-(s+j\omega)t} dt = \left[-B \frac{e^{-(s+j\omega)t}}{s+j\omega} \right]_0^{\infty} = \frac{B}{s+j\omega}$

Time domain	$f(t)$	$u(t)$	$e^{-at}u(t)$	$Ae^{-at}u(t)$	$Be^{-j\omega t}u(t)$
s domain	$F(s)$	$\frac{1}{s}$	$\frac{1}{s+a}$	$\frac{A}{s+a}$	$\frac{B}{s+j\omega}$

13.3 Laplace transform (h)

- Example 13.02

Example 13.02: determine the Laplace transforms of $\sin\omega t$ and $\cos\omega t$.

Solution I:
$$\begin{cases} e^{j\omega t} = \cos\omega t + j\sin\omega t \\ e^{-j\omega t} = \cos\omega t - j\sin\omega t \end{cases} \Rightarrow \begin{cases} \cos\omega t = (e^{j\omega t} + e^{-j\omega t})/2 & (1) \\ \sin\omega t = (e^{j\omega t} - e^{-j\omega t})/j2 & (2) \end{cases}$$

$$\begin{cases} \mathcal{L}(\cos\omega t) = \frac{1}{2}\mathcal{L}(e^{j\omega t} + e^{-j\omega t}) = \frac{1}{2}\mathcal{L}(e^{j\omega t}) + \frac{1}{2}\mathcal{L}(e^{-j\omega t}) = \frac{1}{2}\left(\frac{1}{s - j\omega}\right) + \frac{1}{2}\left(\frac{1}{s + j\omega}\right) = \frac{s}{s^2 + \omega^2} \\ \mathcal{L}(\sin\omega t) = \frac{1}{j2}\mathcal{L}(e^{j\omega t} - e^{-j\omega t}) = \frac{1}{j2}\mathcal{L}(e^{j\omega t}) - \frac{1}{j2}\mathcal{L}(e^{-j\omega t}) = \frac{1}{j2}\left(\frac{1}{s - j\omega}\right) - \frac{1}{j2}\left(\frac{1}{s + j\omega}\right) = \frac{\omega}{s^2 + \omega^2} \end{cases}$$

Therefore,
$$\mathcal{L}(\cos\omega t) = \frac{s}{s^2 + \omega^2}, \quad \mathcal{L}(\sin\omega t) = \frac{\omega}{s^2 + \omega^2}$$

13.3 Laplace transform (i)

- Example 13.02

Example 13.02: determine the Laplace transforms of $\sin\omega t$ and $\cos\omega t$.

Solution II: As $\mathcal{L}(e^{at}) = \frac{1}{s-a}$, $\mathcal{L}(e^{-j\omega t}) = \frac{1}{s+j\omega}$

$$\mathcal{L}(e^{-j\omega t}) = \frac{1}{s+j\omega} = \frac{1}{s+j\omega} \frac{s-j\omega}{s-j\omega} = \frac{s-j\omega}{s^2+\omega^2} = \frac{s}{s^2+\omega^2} - j \frac{\omega}{s^2+\omega^2} \quad (1)$$

$$\mathcal{L}(e^{-j\omega t}) = \mathcal{L}(\cos\omega t - j\sin\omega t) = \mathcal{L}(\cos\omega t) - j\mathcal{L}(\sin\omega t) \quad (2)$$

Comparing (1) and (2): $\mathcal{L}(\cos\omega t) = \frac{s}{s^2+\omega^2}$, $\mathcal{L}(\sin\omega t) = \frac{\omega}{s^2+\omega^2}$

13.3 Laplace transform (j)

- Example 13.03

Example 13.03: determine the Laplace transforms of (a) $r(t) = tu(t)$, that is the ramp function and (b) $f(t) = t^n$.

Solution:

$$(a) F(s) = \mathcal{L}(tu(t)) = \int_{0^-}^{\infty} te^{-st} dt = \left[t \frac{-e^{-st}}{s} \right]_0^{\infty} + \frac{1}{s} \int_{0^-}^{\infty} e^{-st} dt = 0 + 0 - \left(-\frac{1}{s^2} \right) = \frac{1}{s^2}$$

Integration by parts: $\int_a^b u(t)v'(t) dt = u(t)v(t) \Big|_a^b - \int_a^b v(t)u'(t) dt$

with $u(t) = t$, $v'(t) = e^{-st}$,

13.3 Laplace transform (k)

- Example 13.03

Example 13.03: determine the Laplace transforms of (a) $r(t) = tu(t)$, that is the ramp function and (b) $f(t) = t^n$.

Solution:

$$\begin{aligned} (b) \mathcal{L}(t^n) &= \int_{0^-}^{\infty} t^n e^{-st} dt = \left[t^n \frac{-e^{-st}}{s} \right]_{0^-}^{\infty} - \int_{0^-}^{\infty} nt^{n-1} \frac{-e^{-st}}{s} dt \\ &= 0 + \frac{n}{s} \int_{0^-}^{\infty} t^{n-1} e^{-st} dt = \frac{n}{s} \mathcal{L}(t^{n-1}) \end{aligned}$$

So we get a recursive relation: $\mathcal{L}(t^n) = \frac{n}{s} \mathcal{L}(t^{n-1})$, $\forall n$

Therefore, $\mathcal{L}(t^n) = \frac{n}{s} \mathcal{L}(t^{n-1})$, $\mathcal{L}(t^{n-1}) = \frac{n-1}{s} \mathcal{L}(t^{n-2})$, $\mathcal{L}(t^{n-2}) = \frac{n-2}{s} \mathcal{L}(t^{n-3})$, ...

13.3 Laplace transform (I)

- Example 13.03

Example 13.03: determine the Laplace transforms of (a) $r(t) = tu(t)$, that is the ramp function and (b) $f(t) = t^n$.

Solution:

$$\begin{aligned}\mathcal{L}(t^n) &= \frac{n}{s} \mathcal{L}(t^{n-1}) = \frac{n}{s} \frac{n-1}{s} \mathcal{L}(t^{n-2}) = \frac{n}{s} \frac{n-1}{s} \frac{n-2}{s} \mathcal{L}(t^{n-3}) \\ &= \dots = \frac{n}{s} \frac{n-1}{s} \frac{n-2}{s} \frac{n-3}{s} \dots \frac{1}{s} \mathcal{L}(1) = \frac{n!}{s^n} \mathcal{L}(1) = \frac{n!}{s^n} \frac{1}{s} \\ &= \frac{n!}{s^{n+1}} \mathcal{L}(1) = \frac{n!}{s^{n+1}} \frac{1}{s} \\ \Rightarrow \mathcal{L}(t^n) &= \frac{n!}{s^{n+1}}\end{aligned}$$

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13.4 Properties of Laplace transform (a)

- Linearity and scaling properties:

1. Linearity (线性性质)

$$\mathcal{L}[a_1 f_1(t) + a_2 f_2(t)] = a_1 F_1(s) + a_2 F_2(s)$$

2. Scaling (尺度变换性质)

$$\mathcal{L}[f(at)] = \frac{1}{a} F\left(\frac{s}{a}\right)$$

$$\mathcal{L}(f(t)) = \mathcal{L}(e^t) = F(s) = \frac{1}{s-1}$$

$$\mathcal{L}(f(at)) = \frac{1}{a} F\left(\frac{s}{a}\right) = \frac{1}{a} \frac{1}{\left(\frac{s}{a}\right) - 1} = \frac{1}{s-a} = \mathcal{L}(e^{at})$$

13.4 Properties of Laplace transform (b)

- Time and frequency shift:

3. Time shift (时域变换性质)

$$\mathcal{L}[f(t - a)u(t - a)] = e^{-as}F(s)$$

$$\mathcal{L}[\cos \omega(t - a)u(t - a)] = e^{-as} \frac{s}{s^2 + \omega^2}$$

4. Freq. shift (频域变换性质)

$$\mathcal{L}[e^{-at}f(t)u(t)] = F(s + a)$$

$$\mathcal{L}[e^{-at} \cos \omega t u(t)] = \frac{s + a}{(s + a)^2 + \omega^2} \quad \mathcal{L}[e^{-at} \sin \omega t u(t)] = \frac{\omega}{(s + a)^2 + \omega^2}$$

13.4 Properties of Laplace transform (c)

- Differential in time domain:

5. Differential in time domain 时域微分

$$\mathcal{L}[f'(t)] = sF(s) - f(0^-)$$

$$\mathcal{L}[f''(t)] = s^2F(s) - sf(0^-) - f'(0^-)$$

1st order differentiation

$$\begin{aligned}\mathcal{L}\left[\frac{df}{dt}u(t)\right] &= f(t)e^{-st}\Big|_{0^-}^{\infty} - \int_{0^-}^{\infty} f(t)[-se^{-st}] dt \\ &= 0 - f(0^-) + s \int_{0^-}^{\infty} f(t)e^{-st} dt = sF(s) - f(0^-)\end{aligned}$$

Higher-order differentiation

$$\begin{aligned}\mathcal{L}\left\{\frac{d^n f(t)}{dt^n}\right\} &= s^n F(s) - s^{n-1}f(0^-) - s^{n-2}\frac{df(0^-)}{dt} \\ &\quad - s^{n-3}\frac{d^2f(0^-)}{dt^2} - \dots - \frac{d^{n-1}f(0^-)}{dt^{n-1}}.\end{aligned}$$

$$\begin{aligned}\mathcal{L}\left[\frac{d^2f}{dt^2}\right] &= s\mathcal{L}[f'(t)] - f'(0^-) = s[sF(s) - f(0^-)] - f'(0^-) \\ &= s^2F(s) - sf(0^-) - f'(0^-)\end{aligned}$$

2nd order differentiation

13.4 Properties of Laplace transform (d)

- Differential in time domain

6. Integration in time domain 时域积分

$$\mathcal{L}\left[\int_0^t f(x)dx\right] = \frac{1}{s}F(s)$$

$$\mathcal{L}\left[\int_0^t f(x)dx\right] = \int_{0^-}^{\infty} \left[\int_0^t f(x)dx\right] e^{-st} dt = \left[\int_0^t f(x) dx\right] \left(-\frac{1}{s}e^{-st}\right) \Big|_{0^-}^{\infty} - \int_{0^-}^{\infty} \left(-\frac{1}{s}\right) e^{-st} f(t) dt$$

$$\mathcal{L}\left[\int_0^t f(x) dx\right] = \frac{1}{s} \int_{0^-}^{\infty} f(t) e^{-st} dt = \frac{1}{s}F(s)$$

An example:

$$\mathcal{L}(u(t)) = \frac{1}{s} \Rightarrow \mathcal{L}\left[\int_0^t u(t) dt\right] = \mathcal{L}[t] = \frac{1}{s} \left(\frac{1}{s}\right) = \frac{1}{s^2} \quad \mathcal{L}[t^2] = \frac{2}{s^3}$$

13.4 Properties of Laplace transform (e)

- The initial-value theorem (初值定理):

7. Initial-value theorem:

$$f(0) = \lim_{s \rightarrow \infty} sF(s)$$

We know that:

$$sF(s) - f(0) = \mathcal{L}\left[\frac{df}{dt}\right] = \int_{0^-}^{\infty} \frac{df}{dt} e^{-st} dt$$

Let $s \rightarrow \infty$, the integral $\rightarrow 0$, then we have

$$\lim_{s \rightarrow \infty} [sF(s) - f(0)] = 0$$

$$\Rightarrow f(0) = \lim_{s \rightarrow \infty} sF(s)$$

An example:

$$f(t) = e^{-2t} \cos 10t \iff F(s) = \frac{s + 2}{(s + 2)^2 + 10^2}$$

$$\begin{aligned} f(0) &= \lim_{s \rightarrow \infty} sF(s) = \lim_{s \rightarrow \infty} \frac{s^2 + 2s}{s^2 + 4s + 104} \\ &= \lim_{s \rightarrow \infty} \frac{1 + 2/s}{1 + 4/s + 104/s^2} = 1 \end{aligned}$$

13.4 Properties of Laplace transform (f)

- The final-value theorem (终值定理):

8. Final-value theorem:

$$f(\infty) = \lim_{s \rightarrow 0} sF(s)$$

$$f(\infty) = \lim_{s \rightarrow 0} sF(s)$$

The final-value theorem is useful only if $f(\infty)$ exists. This condition is true only if all the poles of $F(s)$, except for a simple pole at the origin, lie in the left half of the s plane. Read pages 685-686 on the textbook as well as Seciton 12.9 on Nillson's book.

9. Freq. domain differentiation 频域微分

$$\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n F(s)}{ds^n}$$

$$\mathcal{L}[te^{-at}u(t)] = -\frac{d}{ds}\left(\frac{1}{s+a}\right) = \frac{1}{(s+a)^2}$$

13.4 Properties of Laplace transform (g)

- **Properties:**

1. **Linearity** $\mathcal{L}[a_1 f_1(t) + a_2 f_2(t)] = a_1 F_1(s) + a_2 F_2(s)$ 2. **Scaling** $\mathcal{L}[f(at)] = \frac{1}{a} F\left(\frac{s}{a}\right)$

3. **Time shift** $\mathcal{L}[f(t - a)u(t - a)] = e^{-as} F(s)$

4. **Freq. shift** $\mathcal{L}[e^{-at} f(t)u(t)] = F(s + a)$

5. **Differential in time domain** $\mathcal{L}[f'(t)] = sF(s) - f(0^-)$

$$\mathcal{L}[f''(t)] = s^2 F(s) - sf(0^-) - f'(0^-)$$

7. **Initial-value theorem:**

$$f(0) = \lim_{s \rightarrow \infty} sF(s)$$

6. **Integration in time domain** $\mathcal{L}\left[\int_0^t f(x)dx\right] = \frac{1}{s} F(s)$

8. **Final-value theorem:**

$$f(\infty) = \lim_{s \rightarrow 0} sF(s)$$

9. **Freq. domain differentiation** $\mathcal{L}[t^n f(t)] = (-1)^n \frac{d^n F(s)}{ds^n}$

13.4 Properties of Laplace transform (h)

- Example 13.05

Example 13.05: determine the Laplace transforms of $f(t) = t^2 \sin(2t)u(t)$.

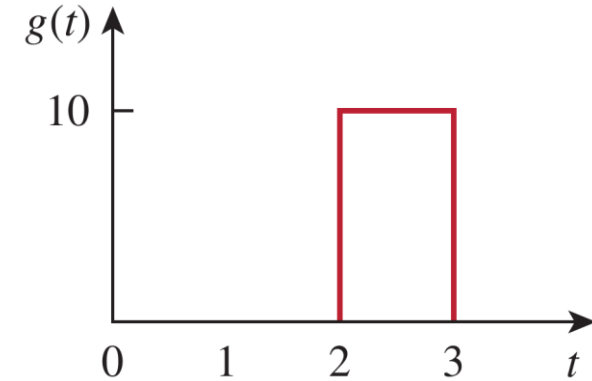
Solution: $\mathcal{L}(\sin(2t)) = \frac{2}{s^2 + 4}$, and $\mathcal{L}(t^n f(t)) = (-1)^n \frac{d^n}{ds^n} F(s)$

$$\begin{aligned} F(s) &= \mathcal{L}(t^2 \sin(2t)) = (-1)^2 2 \frac{d^2}{ds^2} (s^2 + 4)^{-1} = -4 \frac{d}{ds} (s^2 + 4)^{-2} s \\ &= 16s^2 (s^2 + 4)^{-3} - 4(s^2 + 4)^{-2} = \frac{16s^2}{(s^2 + 4)^3} - \frac{4(s^2 + 4)}{(s^2 + 4)^3} \\ &= \frac{12s^2 - 16}{(s^2 + 4)^3} \end{aligned}$$

13.4 Properties of Laplace transform (i)

- Example 13.06

Example 13.06: find the Laplace transform of the gate function in the following figure.



Solution: According to the figure, the gate function can be written as below:

$$g(t) = 10u(t - 2) - 10u(t - 3)$$

$$\mathcal{L}(g(t)) = \mathcal{L}(10u(t - 2)) - \mathcal{L}(10u(t - 3))$$

$$\mathcal{L}[f(t - a)u(t - a)] = e^{-as}F(s)$$

$$\mathcal{L}(g(t)) = \frac{10e^{-2s}}{s} - \frac{10e^{-3s}}{s} = \frac{10}{s}(e^{-2s} - e^{-3s})$$

13.4 Properties of Laplace transform (j)

- Example 13.07

Example 13.07: Find the initial and final values of the function whose Laplace transform is given by:

$$H(s) = \frac{20}{(s+3)(s^2+8s+25)}$$

Solution: (1) To find the initial value, we apply the initial – value theorem:

$$\begin{aligned} h(0) &= \lim_{s \rightarrow \infty} sH(s) = \lim_{s \rightarrow \infty} \frac{20s}{(s+3)(s^2+8s+25)} \\ &= \lim_{s \rightarrow \infty} \frac{\frac{20s}{s^3}}{\left[\frac{(s+3)}{s}\right] \left[\frac{s^2+8s+25}{s^2}\right]} = \lim_{s \rightarrow \infty} \frac{\frac{20}{s^2}}{\left[1 + \frac{3}{s}\right] \left[1 + \frac{8}{s} + \frac{25}{s^2}\right]} = 0 \end{aligned}$$

13.4 Properties of Laplace transform (k)

- Example 13.07

Example 13.07: Find the initial and final values of the function whose Laplace transform is given by:

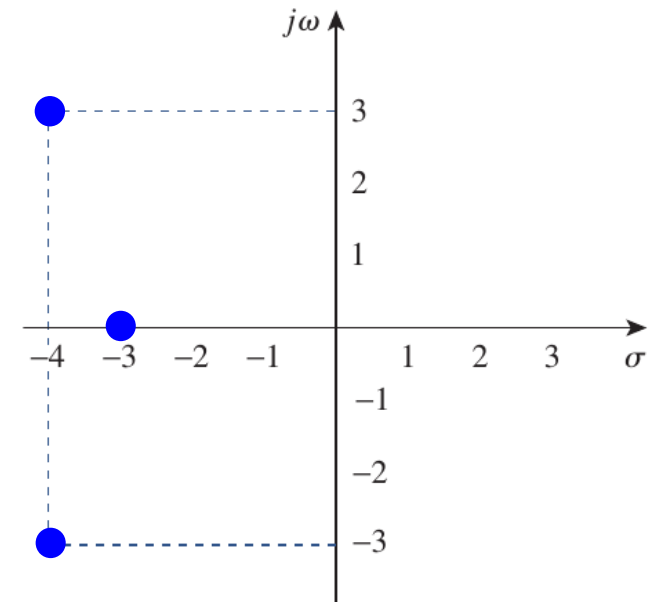
$$H(s) = \frac{20}{(s + 3)(s^2 + 8s + 25)}$$

Solution: (2) To find the final value, we first check whether the final-value theorem is applicable :

$$H(s) = \frac{20}{(s + 3)(s^2 + 8s + 25)} = \frac{20}{(s + 3)(s + 4 - j3)(s + 4 + j3)}$$

There are three poles, $s = -3, 4 - j3, 4 + j3$, which are all on the left half of the s plane

So the final-value theorem can be applied.



13.4 Properties of Laplace transform (I)

- Example 13.07

Example 13.07: Find the initial and final values of the function whose Laplace transform is given by:

$$H(s) = \frac{20}{(s + 3)(s^2 + 8s + 25)}$$

Solution: (3) Now we applied the final-value theorem:

$$h(\infty) = \lim_{s \rightarrow 0} sH(s) = \lim_{s \rightarrow 0} \frac{20s}{(s + 3)(s^2 + 8s + 25)} = 0$$

Check the poles before applying the final-value theorem!

13.5 Time domain to s domain (a)

- Laplace transform pairs (Textbook Page 687):

TABLE 15.2

Laplace transform pairs.*

$f(t)$	$F(s)$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
e^{-at}	$\frac{1}{s + a}$
t	$\frac{1}{s^2}$
t^n	$\frac{n!}{s^{n+1}}$

TABLE 15.2

Laplace transform pairs.*

$f(t)$	$F(s)$
te^{-at}	$\frac{1}{(s + a)^2}$
$t^n e^{-at}$	$\frac{n!}{(s + a)^{n+1}}$
$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$

TABLE 15.2

Laplace transform pairs.*

$f(t)$	$F(s)$
$\sin(\omega t + \theta)$	$\frac{s \sin \theta + \omega \cos \theta}{s^2 + \omega^2}$
$\cos(\omega t + \theta)$	$\frac{s \cos \theta - \omega \sin \theta}{s^2 + \omega^2}$
$e^{-at} \sin \omega t$	$\frac{\omega}{(s + a)^2 + \omega^2}$
$e^{-at} \cos \omega t$	$\frac{s + a}{(s + a)^2 + \omega^2}$

13.5 Time domain to s domain (b)

• Properties of Laplace transform (Textbook Page 687):

TABLE 15.1

Properties of the Laplace transform.

Property	$f(t)$	$F(s)$
Linearity	$a_1 f_1(t) + a_2 f_2(t)$	$a_1 F_1(s) + a_2 F_2(s)$
Scaling	$f(at)$	$\frac{1}{a} F\left(\frac{s}{a}\right)$
Time shift	$f(t - a)u(t - a)$	$e^{-as} F(s)$
Frequency shift	$e^{-at} f(t)$	$F(s + a)$
Time differentiation	$\frac{df}{dt}$	$sF(s) - f(0^-)$
	$\frac{d^2 f}{dt^2}$	$s^2 F(s) - sf(0^-) - f'(0^-)$
	$\frac{d^3 f}{dt^3}$	$s^3 F(s) - s^2 f(0^-) - sf'(0^-) - f''(0^-)$
	$\frac{d^n f}{dt^n}$	$s^n F(s) - s^{n-1} f(0^-) - s^{n-2} f'(0^-) - \dots - f^{(n-1)}(0^-)$

TABLE 15.1

Properties of the Laplace transform.

Property	$f(t)$	$F(s)$
Time integration	$\int_0^t f(x) dx$	$\frac{1}{s} F(s)$
Frequency differentiation	$tf(t)$	$-\frac{d}{ds} F(s)$
Frequency integration	$\frac{f(t)}{t}$	$\int_s^\infty F(s) ds$
Time periodicity	$f(t) = f(t + nT)$	$\frac{F_1(s)}{1 - e^{-sT}}$
Initial value	$f(0)$	$\lim_{s \rightarrow \infty} sF(s)$
Final value	$f(\infty)$	$\lim_{s \rightarrow 0} sF(s)$
Convolution	$f_1(t) * f_2(t)$	$F_1(s)F_2(s)$

13.5 Time domain to s domain (c)

- How to transform a signal from time domain to s domain

You should know the Laplace transform of some basic signals:

- a) Unit step $F(s) = 1/s$ and unit impulse $F(s) = 1$
- b) Exponential: $f(t) = e^{at} \leftrightarrow F(s) = 1/(s - a)$
- c) Sinusoids: $f(t) = \cos\omega t \leftrightarrow F(s) = s/(s^2 - \omega^2)$ & $f(t) = \sin\omega t \leftrightarrow F(s) = \omega/(s^2 - \omega^2)$

these, combined with the table of Laplace transform and the properties (linearity, scaling, integration/differentiation, ...) given above will get you pretty far.

And don't forget that you can always integrate based on the definition:

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

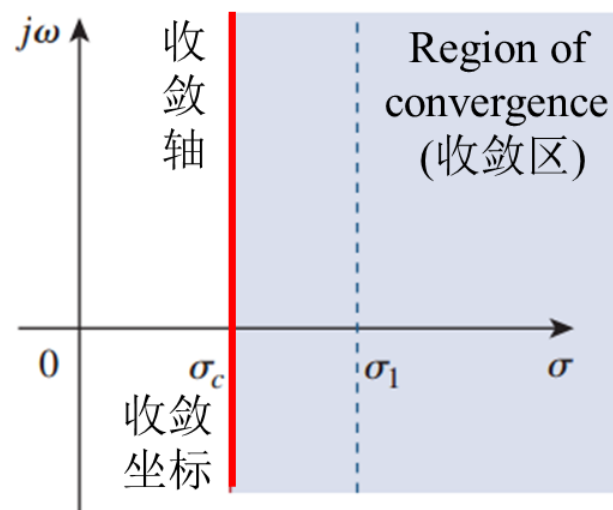
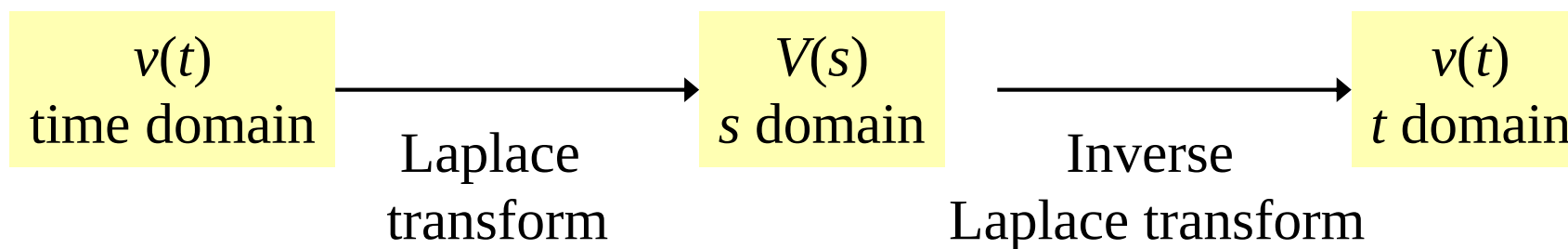
Lesson 13 Introduction to the Laplace transform

13.1-13.2	Introduction	Slides 06-13
13.3	Laplace transform: what is it?	Slides 15-26
13.4-13.5	Properties of Laplace transform	Slides 28-42
13.6	Inverse Laplace transform	Slides 44-67
13.7	*Convolution Integral	Slides 69-75
13.8	Summary	Slides 77-78

13.6 Inverse Laplace transform (a)

- Inverse Laplace transform (拉普拉斯反变换):

We have learned how to transform a signal from time domain to s domain:



$$\mathcal{L}^{-1}[F(s)] = f(t) = \frac{1}{2\pi j} \int_{\sigma_1 - j\infty}^{\sigma_1 + j\infty} F(s) e^{st} ds$$

The direct application this involves some knowledge about complex analysis beyond the scope of this book.

We will rather use the look-up table.

13.6 Inverse Laplace transform (b)

- Example 13.08 (查表法):

Example 13.08: obtain the inverse Laplace transform of $F(s) = \frac{3}{s} - \frac{5}{s+1} + \frac{6}{s^2+4}$

TABLE 15.2

Laplace transform pairs.*

$f(t)$	$F(s)$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
e^{-at}	$\frac{1}{s+a}$
t	$\frac{1}{s^2}$
t^n	$\frac{n!}{s^{n+1}}$
$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$

Solution: (1) Based on the look-up table, it is easy to know:

$$\begin{aligned}\mathcal{L}^{-1}(F(s)) &= \mathcal{L}^{-1}\left(\frac{3}{s} - \frac{5}{s+1} + \frac{6}{s^2+4}\right) \\ &= \mathcal{L}^{-1}\left(\frac{3}{s}\right) - \mathcal{L}^{-1}\left(\frac{5}{s+1}\right) + \mathcal{L}^{-1}\left(\frac{6}{s^2+4}\right) \rightarrow \frac{6}{s^2+4} = \frac{3 \times 2}{s^2+2^2} \\ &= [3 - 5e^{-t} + 3 \sin(2t)]u(t)\end{aligned}$$

Practice and get familiar with the tables.

13.6 Inverse Laplace transform (c)

- Steps to apply inverse Laplace transform:

- a) Based on the table we can inversely transform simple $F(s)$, yet what if it is more complicated?
- b) In circuit analysis, the Laplace transform encountered **generally has a form of $F(s) = N(s)/D(s)$** , in which $N(s)$ is the numerator polynomial and $D(s)$ is the denominator polynomial. **Roots of $N(s)=0$ and $D(s)=0$ are called zeros and poles of $F(s)$, respectively.**

$$F(s) = \frac{N(s)}{D(s)} = \frac{N(s)}{(s + p_1)(s + p_1) \cdots (s + p_n)} \quad \Rightarrow \quad F(s) = \frac{k_1}{s + p_1} + \frac{k_2}{s + p_2} + \cdots + \frac{k_n}{s + p_n}$$

Step 1: decompose $F(s)$ into simple terms using partial fraction expansion.
(部分分式展开)

Step 2: find the inverse of each term by matching entries in Table 15.2
(查表得出时域信号)

13.6 Inverse Laplace transform (d)

- Steps to apply inverse Laplace transform:

Possible forms

$$F(s) = \frac{N(s)}{D(s)} = \frac{a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}$$

1. When $m > n$, $F(s)$ is a proper rational function (有理真分式)

The roots of $D(s)$
are or include

- Distinct real roots (独立实根)
- Repeated real roots (实重根)
- Distinct complex roots (不等复根)
- Repeat complex roots (多重复根)

2. When $m \leq n$, $F(s)$ is a proper rational function (有理假分式)

Methods

部分分式分解方法

Residue method
(留数法)

Algebraic method
(代数法)

13.6 Inverse Laplace transform (e)

- Distinct real roots of $D(s)$ | $D(s)$ 具有独立实根:

- a) When the roots of $D(s)$ are distinct real numbers, $F(x)$ can be expanded as follows.
- b) Here the expansion coefficients k_1 , k_2 and k_3 are known as the residues (留数) of $F(x)$. There are many ways to find such coefficients and one of them the residue method (留数法).

$$F(s) = \frac{N(s)}{D(s)} = \frac{96 \times (s + 5)(s + 12)}{s(s + 8)(s + 6)} \equiv \frac{k_1}{s} + \frac{k_2}{s + 8} + \frac{k_3}{s + 6}$$

(1) To obtain k_1 , multiply both sides by s and then let $s = 0$:

$$\frac{96 \times (s + 5)(s + 12)}{(s + 8)(s + 6)} \equiv k_1 + \frac{sk_2}{s + 8} + \frac{sk_3}{s + 6} \Rightarrow \frac{96 \times 5 \times 12}{8 \times 6} \equiv k_1 \Rightarrow k_1 = 120$$

13.6 Inverse Laplace transform (f)

- Distinct real roots of $D(s)$ | $D(s)$ 具有独立实根:

- a) When the roots of $D(s)$ are distinct real numbers, $F(x)$ can be expanded as follows.
- b) Here the expansion coefficients k_1 , k_2 and k_3 are known as the residues (留数) of $F(x)$. There are many ways to find such coefficients and one of them the residue method (留数法).

(2) Multiply both sides by $s + 8$ and then let $s = -8 \Rightarrow k_2 = -72$

$$\frac{96 \times (s + 5)(s + 12)}{s(s + 6)} = \frac{(s + 8)k_1}{s} + k_2 + \frac{(s + 8)k_3}{s} \Rightarrow k_2 = \frac{96 \times (-3) \times 4}{(-8) \times (-2)} = -72$$

(3) Multiply both sides by $s + 6$ and then let $s = -6 \Rightarrow k_3 = 48$

$$\frac{96 \times (s + 5)(s + 12)}{s(s + 8)} = \frac{(s + 6)k_1}{s} + \frac{(s + 6)k_1}{s + 8} + k_3 \Rightarrow k_3 = \frac{96 \times (-1) \times 6}{(-6) \times 2} = 48$$

13.6 Inverse Laplace transform (g)

- Distinct real roots of $D(s)$ | $D(s)$ 具有独立实根:

- a) When the roots of $D(s)$ are distinct real numbers, $F(x)$ can be expanded as follows.
- b) Here the expansion coefficients k_1 , k_2 and k_3 are known as the residues (留数) of $F(x)$. There are many ways to find such coefficients and one of them the residue method (留数法).

$$F(s) = \frac{N(s)}{D(s)} = \frac{96 \times (s+5)(s+12)}{s(s+8)(s+6)} \equiv \frac{k_1}{s} + \frac{k_2}{s+8} + \frac{k_3}{s+6} = \frac{120}{s} - \frac{72}{s+8} + \frac{48}{s+6}$$

$$\mathcal{L}^{-1}(F(s)) = \mathcal{L}^{-1}\left(\frac{120}{s}\right) - \mathcal{L}^{-1}\left(\frac{72}{s+8}\right) - \mathcal{L}^{-1}\left(\frac{48}{s+6}\right)$$

$$= 120u(t) - 72e^{-8t}u(t) + 48e^{-6t}u(t)$$

$$= (120 - 72e^{-8t} + 48e^{-6t})u(t)$$

13.6 Inverse Laplace transform (h)

- Repeated real roots of $D(s)$ | $D(s)$ 具有实重根:

a) When $D(s)$ has repeated real roots, $F(s)$ can be expanded as follows, and expansion coefficients can be obtained using again the residue method.

$$\frac{100(s + 25)}{s(s + 5)^3} \equiv \frac{k_1}{s} + \frac{k_2}{(s + 5)^3} + \frac{k_3}{(s + 5)^2} + \frac{k_4}{(s + 5)}$$



$$F(s) = \frac{k_n}{(s + p)^n} + \frac{k_{n-1}}{(s + p)^{n-1}} + \cdots + \frac{k_2}{(s + p)^2} + \frac{k_1}{s + p} + F_1(s)$$

Read Page 511, section 15.4.2 on the textbook

13.6 Inverse Laplace transform (i)

- Repeated real roots of $D(s)$ | $D(s)$ 具有实重根:

a) When $D(s)$ has repeated real roots, $F(s)$ can be expanded as follows, and expansion coefficients can be obtained using again the residue method (Read Page 511 section 15.4.2 on the textbook).

$$\frac{100(s + 25)}{s(s + 5)^3} \equiv \frac{k_1}{s} + \frac{k_2}{(s + 5)^3} + \frac{k_3}{(s + 5)^2} + \frac{k_4}{(s + 5)}$$

(1) To obtain k_1 , multiply both sides by s and then let $s = 0$:

$$\frac{2500}{5^3} \equiv k_1 \Rightarrow k_1 = 20$$

(2) To obtain k_2 , multiply both sides by $(s + 5)^3$ and then let $s + 5 = 0$, i.e. $s = -5$:

$$\frac{2000}{-5} \equiv k_2 \Rightarrow k_2 = -400$$

13.6 Inverse Laplace transform (j)

- Repeated real roots of $D(s)$ | $D(s)$ 具有实重根:

a) When $D(s)$ has repeated real roots, $F(x)$ can be expanded as follows, and expansion coefficients can be obtained using again the residue method (Read Page 511 section 15.4.2 on the textbook).

$$\frac{100(s + 25)}{s(s + 5)^3} \equiv \frac{k_1}{s} + \frac{k_2}{(s + 5)^3} + \frac{k_3}{(s + 5)^2} + \frac{k_4}{(s + 5)}$$

(3) To obtain k_3 , multiply both sides by $(s + 5)^3$ and differentiate, then let $s + 5 = 0$, i.e. $s = -5$:

Multiplication:
$$\frac{100(s + 25)}{s} = \frac{k_1(s + 5)^3}{s} + k_2 + k_3(s + 5) + k_4(s + 5)^2$$

Differentiation:
$$\frac{d}{ds} \frac{100(s + 25)}{s} = \frac{d}{ds} \frac{k_1(s + 5)^3}{s} + \frac{d}{ds} k_2 + \frac{d}{ds} k_3(s + 5) + \frac{d}{ds} k_4(s + 5)^2$$

$$\Rightarrow \frac{s - (s + 25)}{s^2} = k_3 \quad \text{Let } s = -5 \Rightarrow k_3 = -100$$

13.6 Inverse Laplace transform (k)

- Repeated real roots of $D(s)$ | $D(s)$ 具有实重根:

a) When $D(s)$ has repeated real roots, $F(x)$ can be expanded as follows, and expansion coefficients can be obtained using again the residue method (Read Page 511 section 15.4.2 on the textbook).

(4) To obtain k_4 , multiply both sides by $(s + 5)^3$ and differentiate, then let $s + 5 = 0$, i.e. $s = -5$:

$$\frac{d^2}{ds^2} \frac{100(s + 25)}{s} = \frac{d^2}{ds^2} \frac{k_1(s + 5)^3}{s} + \frac{d^2}{ds^2} k_2 + \frac{d^2}{ds^2} k_3(s + 5) + \frac{d^2}{ds^2} k_4(s + 5)^2 \Rightarrow k_4 = -20$$

(5) Now we can transform $F(s)$ back into time domain:

$$F(s) = \frac{20}{s} - \frac{400}{(s + 5)^3} - \frac{100}{(s + 5)^2} - \frac{20}{(s + 5)}$$

$$\mathcal{L}^{-1} \left(\frac{k}{(s + a)^r} \right) = \frac{kt^{r-1}e^{-at}}{(r-1)!} u(t)$$

$$f(t) = \mathcal{L}^{-1}(F(x)) = [20 - 200t^2e^{-5t} - 100te^{-5t} - 20e^{-5t}]u(t)$$

13.6 Inverse Laplace transform (I)

- Distinct complex roots of $D(s)$ | $D(s)$ 具有不等的复数根:

a) When $D(s)$ has **distinct complex roots**, $F(s)$ can be expanded as follows, and expansion coefficients can be obtained using again the residue method.

$$F(s) = \frac{100(s+3)}{(s+6)(s^2+6s+25)} = \frac{100(s+3)}{(s+6)(s+3-j4)(s+3+j4)} = \frac{k_1}{s+6} + \frac{k_2}{s+3-j4} + \frac{k_3}{s+3+j4}$$

(1) To obtain k_1 , multiply both sides by $s+6$ and then let $s = -6$:

$$k_1 = \frac{-300}{25} = -12$$

(2) To obtain k_2 , multiply both sides by $s+3-j4$ and then let $s = -3+j4$:

$$k_2 = 6 - j8 = 10e^{-j53.13^\circ}$$

(3) To obtain k_3 , multiply both sides by $s+3+j4$ and then let $s = -3-j4$:

$$k_3 = 6 + j8 = 10e^{j53.13^\circ}$$

13.6 Inverse Laplace transform (m)

- Distinct complex roots of $D(s)$ | $D(s)$ 具有不等的复数根:

a) When $D(s)$ has **distinct complex roots**, $F(s)$ can be expanded as follows, and expansion coefficients can be obtained using again the residue method.

$$F(s) = \frac{-12}{s+6} + \frac{10e^{-j53.13^\circ}}{s+3-j4} + \frac{10e^{j53.13^\circ}}{s+3+j4}$$

$$f(t) = \mathcal{L}^{-1}(F(s)) = u(t)[-12e^{-6t} + 10e^{-j53.13^\circ}e^{-(3-j4)t} + 10e^{j53.13^\circ}e^{-(3+j4)t}]$$

$$\begin{aligned} &= 10e^{-j53.13^\circ}e^{-3t}e^{j4t} + 10e^{j53.13^\circ}e^{-3t}e^{-j4t} = 10e^{-3t}[e^{j(4t-53.13^\circ)} + e^{-j(4t-53.13^\circ)}] \\ &= 20e^{-3t}\cos(4t - 53.13^\circ) \end{aligned}$$

$$\text{Therefore, } f(t) = u(t)[-12e^{-6t} + 20e^{-3t}\cos(4t - 53.13^\circ)]$$

13.6 Inverse Laplace transform (n)

- Distinct complex roots of $D(s)$ | $D(s)$ 具有不等的复数根:

- When analyzing lumped linear circuits, it's often to have distinct complex roots.
- If the $D(s)$ includes factors of the form $s + \alpha - j\beta$ and $s + \alpha + j\beta$, then in the partial fraction expansion we often observe terms of:

$$\frac{k}{s + \alpha - j\beta} + \frac{k^*}{s + \alpha + j\beta} \quad (k = |k|e^{j\theta} = |k|\angle\theta \quad \text{and} \quad k^* = |k|e^{-j\theta} = |k|\angle(-\theta))$$

$$\mathcal{L}^{-1}\left(\frac{k}{s + \alpha - j\beta} + \frac{k^*}{s + \alpha + j\beta}\right) = 2|k|e^{-\alpha t} \cos(\beta t + \theta) u(t)$$

(Note that k for $s + \alpha - j\beta$ and k^* for $s + \alpha + j\beta$)

13.6 Inverse Laplace transform (o)

- Repeated complex roots of $D(s)$ | $D(s)$ 具有多重复根:

a) When $D(s)$ has repeated complex roots, $F(s)$ can be expanded as follows, and expansion coefficients can be obtained using again the residue method.

$$F(s) = \frac{768}{(s + 3 - j4)^2(s + 3 + j4)^2} = \frac{k_1}{(s + 3 - j4)^2} + \frac{k_2}{s + 3 - j4} + \frac{k_1^*}{(s + 3 + j4)^2} + \frac{k_2^*}{s + 3 + j4}$$

(1) To obtain k_1 , multiply both sides by $(s + 3 - j4)^2$ and then let $s = -3 + j4$:

$$k_1 = \frac{768}{(-3 + j4 + 3 + j4)^2} = \frac{768}{(j8)^2} = -12 \quad k_1^* = -12$$

(2) To obtain k_2 , multiply both sides by $(s + 3 - j4)^2$, differentiate, and then let $s = -3 + j4$:

$$k_2 = \left[\frac{d}{ds} \frac{768}{(s + 3 + j4)^2} \right]_{s=-3+j4} = \frac{1536}{(j8)^3} = -j3 = 3\angle(-90^\circ) \quad k_2^* = j3 = 3\angle(90^\circ)$$

13.6 Inverse Laplace transform (p)

- Repeated complex roots of $D(s)$ | $D(s)$ 具有多重复数根:

a) When $D(s)$ has repeated complex roots, $F(s)$ can be expanded as follows, and expansion coefficients can be obtained using again the residue method.

$$F(s) = \frac{-12}{(s+3-j4)^2} + \frac{-12}{(s+3+j4)^2} + \frac{3\angle(-90^\circ)}{s+3-j4} + \frac{3\angle(90^\circ)}{s+3+j4}$$

$$\Rightarrow f(t) = \mathcal{L}^{-1}(F(s)) = [-24te^{-3t}\cos 4t + 6e^{-3t}\cos(4t - 90^\circ)]u(t)$$

$$\mathcal{L}^{-1}\left(\frac{k}{(s+a-j\beta)^r} + \frac{k^*}{(s+a+j\beta)^r}\right) = \frac{2|k|t^{r-1}}{(r-1)!}e^{-at}\cos(\beta t + \theta)u(t)$$

13.6 Inverse Laplace transform (q)

- When $F(x)$ is an improper rational function (假分式):

a) $F(s)$ being an improper rational function won't affect much the inverse Laplace transform, since an improper rational function can be expanded to be the sum of a polynomial and a proper rational function.

$$F(s) = \frac{s^4 + 13s^3 + 66s^2 + 200s + 300}{s^2 + 9s + 20}$$

$$F(s) = s^2 + 4s + 10 + \frac{30s + 100}{s^2 + 9s + 20} = s^2 + 4s + 10 - \frac{20}{s + 4} + \frac{50}{s + 5}$$

$$\Rightarrow f(t) = \mathcal{L}^{-1}(F(s)) = \frac{d^2 \delta(t)}{dt^2} + 4 \frac{d\delta(t)}{dt} + 10\delta(t) - [20e^{-4t} - 50e^{-5t}]u(t)$$

13.6 Inverse Laplace transform (r)

- 3 useful transform pairs:

$$F(s) = \frac{k}{(s + a)^r}$$

$$f(t) = \frac{kt^{r-1}e^{-at}}{(r-1)!}u(t)$$

$$F(s) = \frac{k}{s + \alpha - j\beta} + \frac{k^*}{s + \alpha + j\beta}$$

$$f(t) = 2|k|e^{-\alpha t} \cos(\beta t + \theta) u(t)$$

$$F(s) = \frac{k}{(s + a - j\beta)^r} + \frac{k^*}{(s + a + j\beta)^r}$$

$$f(t) = \frac{2|k|t^{r-1}}{(r-1)!} e^{-at} \cos(\beta t + \theta) u(t)$$

13.6 Inverse Laplace transform (s)

- Example 13.09 (留数法和代数法):

Example 13.09: obtain $v(t)$ if $V(s) = (s^2 + 12)/(s(s + 2)(s + 3))$

Solution I (留数法):

$$V(s) = \frac{s^2 + 12}{s(s + 2)(s + 3)} = \frac{A}{s} + \frac{B}{s + 2} + \frac{C}{s + 3}$$



$$\left\{ \begin{array}{l} A = [sF(s)]_{s=0} = \left[\frac{s^2 + 12}{(s + 2)(s + 3)} \right]_{s=0} = 2 \\ B = [(s + 2)F(s)]_{s=-2} = \left[\frac{s^2 + 12}{s(s + 3)} \right]_{s=-2} = -8 \\ C = [(s + 3)F(s)]_{s=-3} = \left[\frac{s^2 + 12}{s(s + 2)} \right]_{s=-3} = 7 \end{array} \right.$$



$$\begin{aligned} v(t) &= \mathcal{L}^{-1}(V(s)) \\ &= \mathcal{L}^{-1}\left(\frac{2}{s} - \frac{8}{s + 2} + \frac{7}{s + 3}\right) \\ &= (2 - 8e^{-2t} + 7e^{-3t})u(t) \end{aligned}$$

13.6 Inverse Laplace transform (t)

- Example 13.09 (留数法和代数法):

Example 13.09: obtain $v(t)$ if $V(s) = (s^2 + 12)/(s(s + 2)(s + 3))$

Solution II (代数法):

$$V(s) = \frac{s^2 + 12}{s(s + 2)(s + 3)} = \frac{A}{s} + \frac{B}{s + 2} + \frac{C}{s + 3}$$

Multiplying both side by $s(s + 2)(s + 3)$:

$$s^2 + 12 = (s + 2)(s + 3)A + s(s + 3)B + s(s + 2)C$$

$$\Rightarrow s^2 + 12 = A(s^2 + 5s + 6) + B(s^2 + 3s) + C(s^2 + 2s)$$

$$\Rightarrow s^2 + 12 = (A + B + C)s^2 + (5A + 3B + 2C)s + 6A$$

13.6 Inverse Laplace transform (u)

- Example 13.09 (留数法和代数法):

Example 13.09: obtain $v(t)$ if $V(s) = (s^2 + 12)/(s(s + 2)(s + 3))$

Solution II (代数法): $s^2 + 12 = (A + B + C)s^2 + (5A + 3B + 2C)s + 6A$

$$\Rightarrow \begin{cases} A + B + C = 1 \\ 5A + 3B + 2C = 0 \\ 6A = 12 \end{cases} \Rightarrow \begin{cases} A = 2 \\ B = -8 \\ C = 7 \end{cases}$$

$$\Rightarrow v(t) = \mathcal{L}^{-1}\left(\frac{2}{s} - \frac{8}{s+2} + \frac{7}{s+3}\right) = (2 - 8e^{-2t} + 7e^{-3t})u(t)$$

13.6 Inverse Laplace transform (v)

- Example 13.10 (拉普拉斯求解微分方程):

Example 13.10: use Laplace transform to solve the differential equation, $v(0) = 1$ and $v'(0) = -2$.

$$\frac{d^2v(t)}{dt^2} + 6\frac{dv(t)}{dt} + 8v(t) = 2u(t)$$

Solution: (1) $v(t)$ in time domain $\rightarrow V(s)$ in s domain to solve the equation

$$\mathcal{L}\left(\frac{d^2v(t)}{dt^2}\right) + 6\mathcal{L}\left(\frac{dv(t)}{dt}\right) + 6\mathcal{L}(v(t)) = 2\mathcal{L}(u(t))$$

$$s^2V(s) - sV(0) - V'(0) + 6[sV(s) - V(0)] + 6V(s) = \frac{2}{s}$$

$$s^2V(s) - s + 2 + 6sV(s) - 6 + 6V(s) = \frac{2}{s} \quad \Rightarrow \quad V(s) = \frac{s^2 + 4s + 2}{s(s+2)(s+4)}$$

13.6 Inverse Laplace transform (w)

- Example 13.10 (拉普拉斯求解微分方程):

Example 13.10: use Laplace transform the solve the differential equation, $v(0) = 1$ and $v'(0) = -2$.

$$\frac{d^2v(t)}{dt^2} + 6\frac{dv(t)}{dt} + 8v(t) = 2u(t)$$

Solution: (2) $V(s)$ in s domain to find the residues

$$V(s) = \frac{s^2 + 4s + 2}{s(s + 2)(s + 4)} \Rightarrow V(s) = \frac{A}{s} + \frac{B}{s + 2} + \frac{C}{s + 4}$$

$$A = [sF(x)]_{s=0} = \left[\frac{s^2 + 4s + 2}{(s + 2)(s + 4)} \right]_{s=0} = \frac{1}{4} \quad B = [(s + 2)F(x)]_{s=-2} = \left[\frac{s^2 + 4s + 2}{s(s + 4)} \right]_{s=-2} = \frac{1}{2}$$

$$C = [(s + 4)F(x)]_{s=-4} = \left[\frac{s^2 + 4s + 2}{s(s + 2)} \right]_{s=-4} = \frac{1}{4}$$

13.6 Inverse Laplace transform (x)

- Example 13.10 (拉普拉斯求解微分方程):

Example 13.10: use Laplace transform the solve the differential equation, $v(0) = 1$ and $v'(0) = -2$.

$$\frac{d^2v(t)}{dt^2} + 6\frac{dv(t)}{dt} + 8v(t) = 2u(t)$$

Solution: (2) $V(s)$ in s domain in s domain to find the residues

$$V(s) = \frac{s^2 + 4s + 2}{s(s+2)(s+4)} \Rightarrow V(s) = \frac{1/4}{s} + \frac{1/2}{s+2} + \frac{1/4}{s+4}$$

$$v(t) = \mathcal{L}^{-1}(V(s)) = \mathcal{L}^{-1}\left(\frac{1/4}{s} + \frac{1/2}{s+2} + \frac{1/4}{s+4}\right)$$

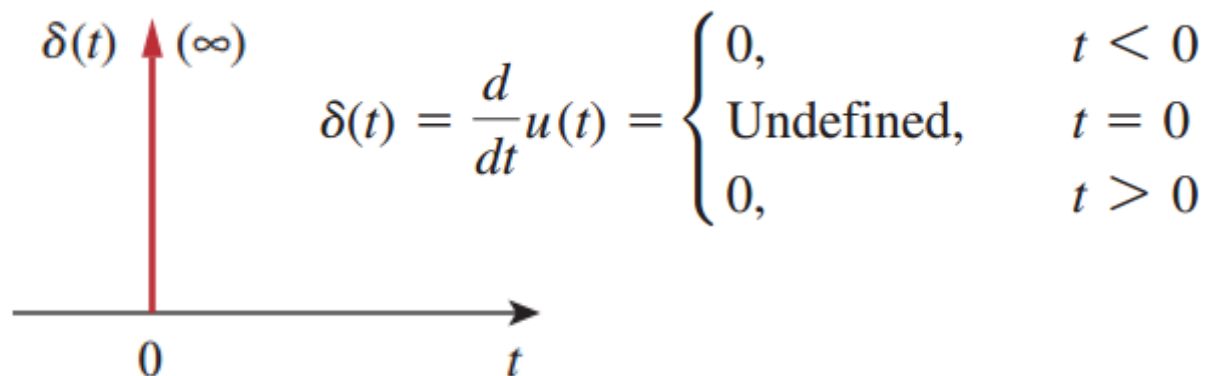
$$\Rightarrow v(t) = \frac{1}{4}(1 + 2e^{-2t} + e^{-4t})u(t)$$

Lesson 13 Introduction to the Laplace transform

13.1-13.2	Introduction	Slides 06-13
13.3	Laplace transform: what is it?	Slides 15-26
13.4-13.5	Properties of Laplace transform	Slides 28-42
13.6	Inverse Laplace transform	Slides 44-67
13.7	*Convolution Integral	Slides 69-75
13.8	Summary	Slides 77-78

13.7 Convolution Integral (a)

- Recall: unit impulse function:



“sampling” property of δ function

$$f(t)\delta(t - t_0) = f(t_0)\delta(t - t_0)$$

$$\begin{aligned} & \int_a^b f(t)\delta(t - t_0)dt \\ &= f(t_0) \int_a^b \delta(t - t_0)dt = f(t_0) \end{aligned}$$

- The unit impulse function $\delta(t - t_0)$ is zero everywhere except at t_0 , where it is undefined.
- The unit area is known as the strength of the impulse function. When an impulse function has a strength other than unity, the area of the impulse is equal to its strength.

13.7 Convolution integral (b)

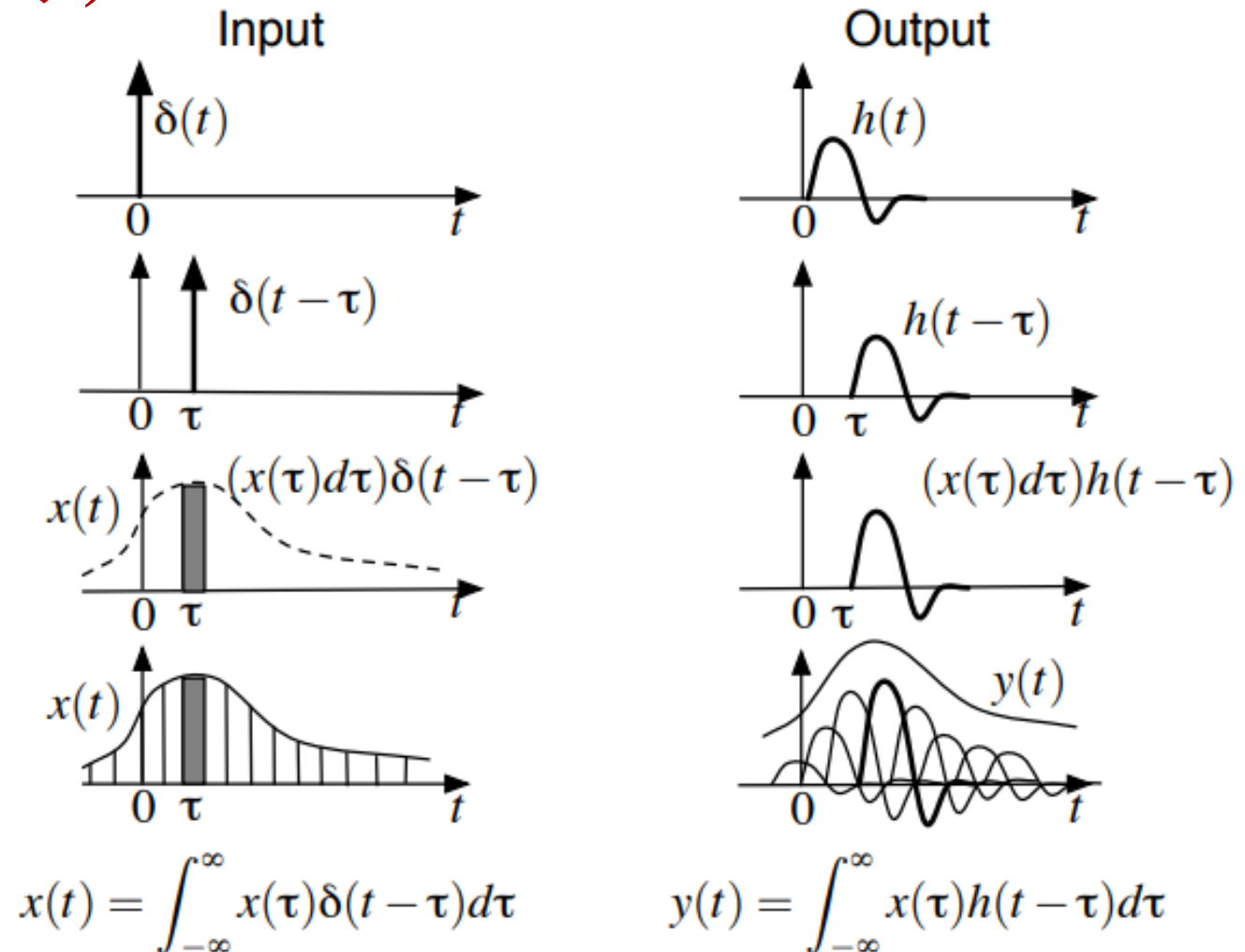
- Convolution integral (卷积积分):

- a) Convolution is a valuable tool to engineers, providing a means of viewing and characterizing physical systems.
- b) It is used in finding the response $y(t)$ of a system to an excitation $x(t)$, knowing the system impulse response $h(t)$.

$$y(t) = \int_{-\infty}^{\infty} x(\lambda)h(t - \lambda) d\lambda$$

$$y(t) = x(t) * h(t)$$

- c) Here λ is a dummy variable (虚拟变量) and the asterisk denotes convolution.

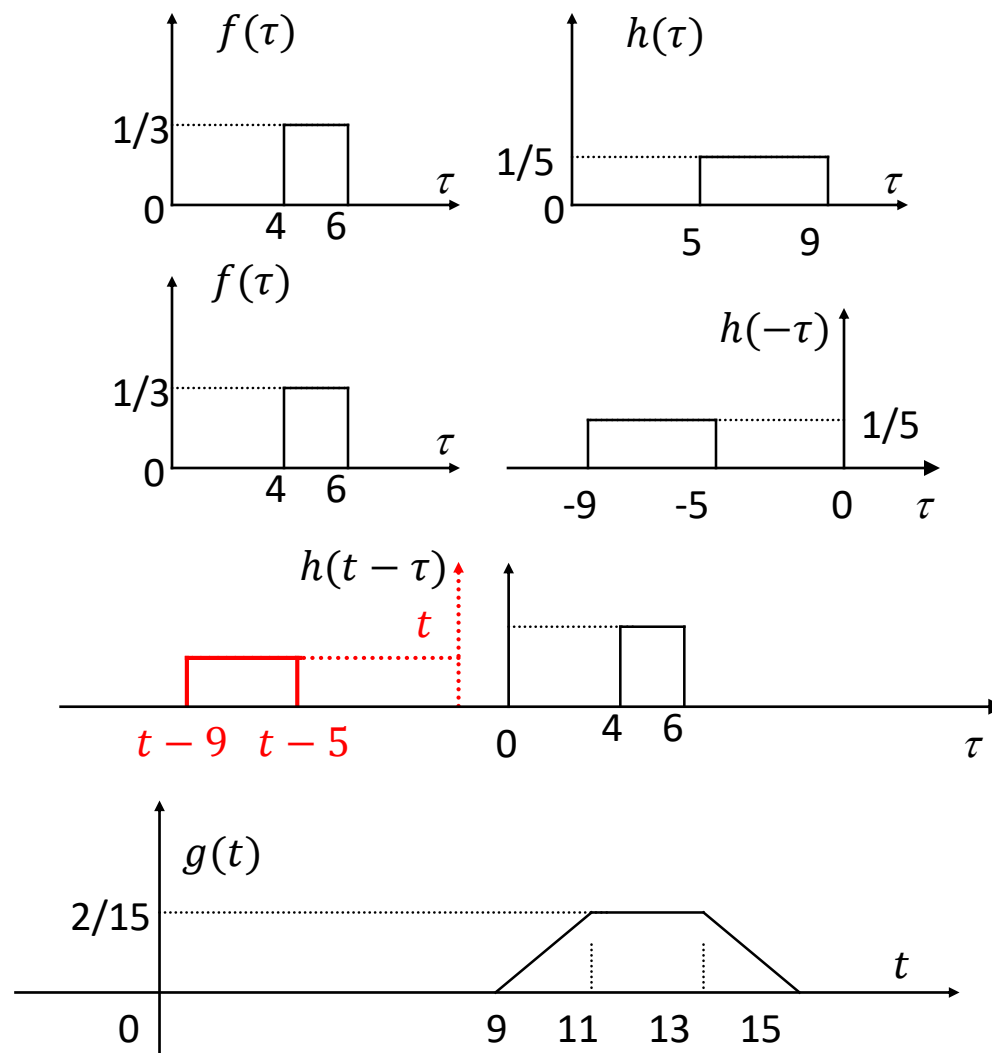


13.7 Convolution integral (c)

- Convolution integral (卷积积分):

Steps:

1. Plot out the two functions to be convoluted $f(\tau)$ and $h(\tau)$.
2. Fold $h(\tau)$ yielding $h(-\tau)$;
3. Shift $h(-\tau)$ to any given position t , yielding $h[-(\tau - t)] = h(t - \tau)$;
4. Multiply with $f(\tau)$.
5. Calculate the area between the resulting curve and t -axis.
6. Repeat for a different t

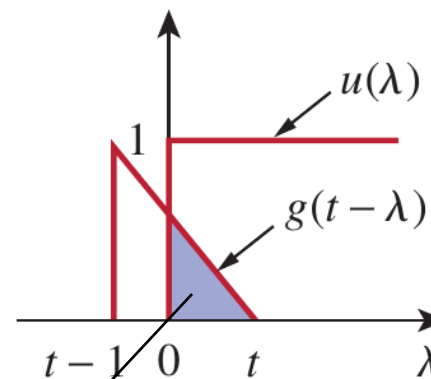
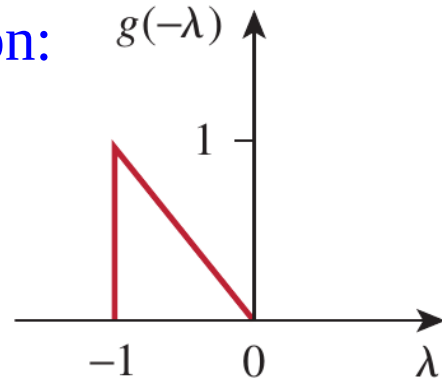
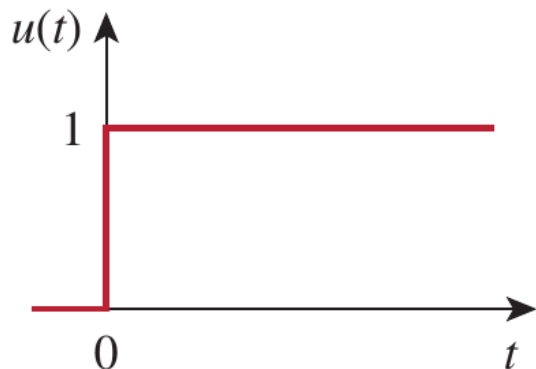
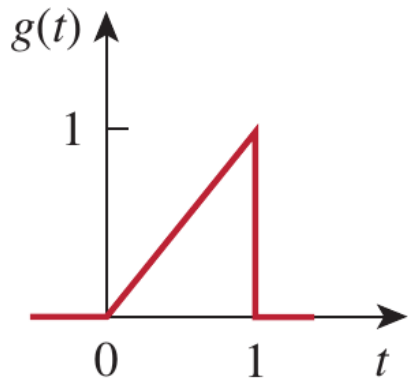


13.7 Convolution integral (d)

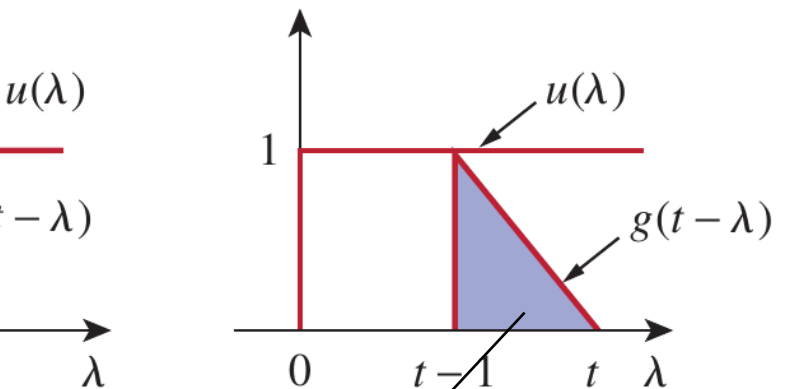
- Example 13.10:

Example 13.10: Graphically convolve $g(t)$ and $u(t)$ shown in the following figure

Solution:



$$\begin{aligned} y(t) &= \int_0^t (1)(t - \lambda) d\lambda = \left(t\lambda - \frac{1}{2}\lambda^2 \right) \Big|_0^t \\ &= t^2 - \frac{t^2}{2} = \frac{t^2}{2}, \quad 0 \leq t \leq 1 \end{aligned}$$

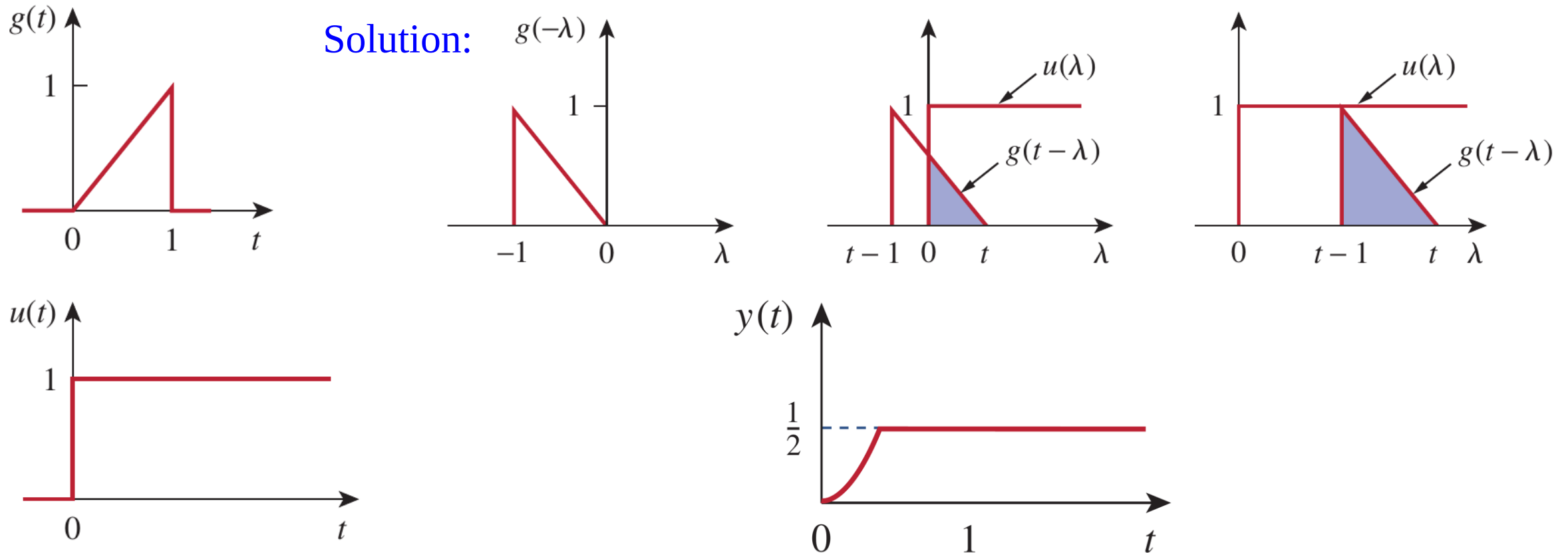


$$\begin{aligned} y(t) &= \int_{t-1}^t (1)(t - \lambda) d\lambda \\ &= \left(t\lambda - \frac{1}{2}\lambda^2 \right) \Big|_{t-1}^t = \frac{1}{2}, \quad t \geq 1 \end{aligned}$$

13.7 Convolution integral (e)

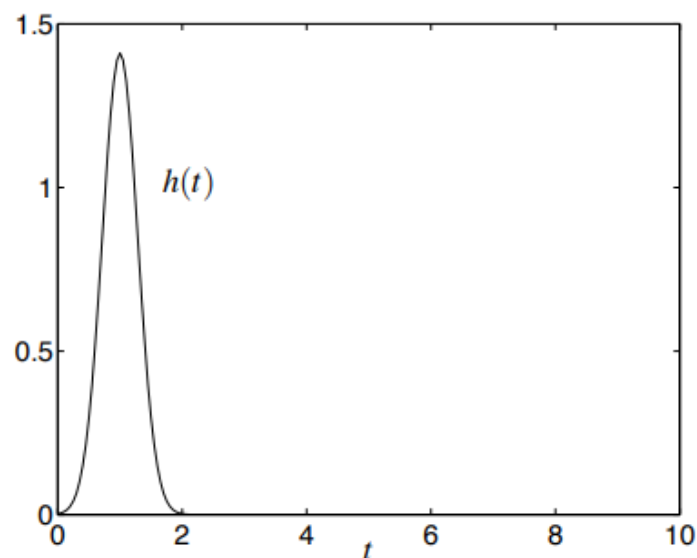
- Example 13.11:

Example 13.11: Graphically convolve $g(t)$ and $u(t)$ shown in the following figure

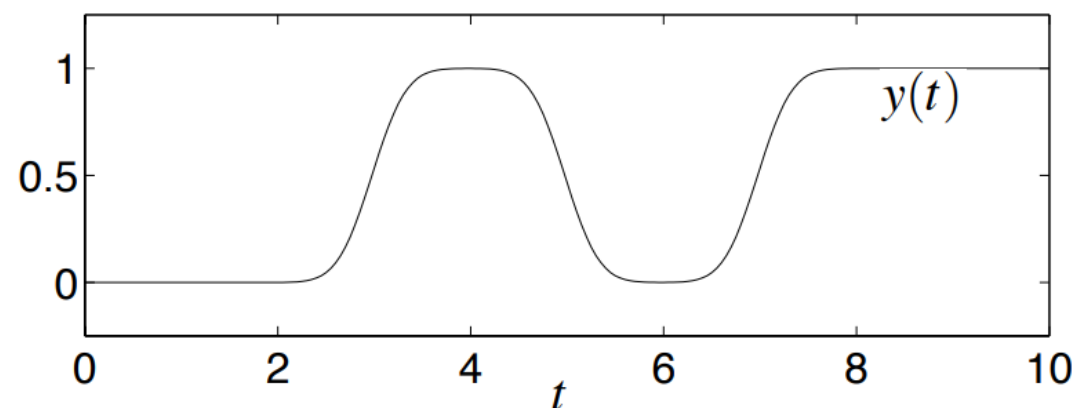
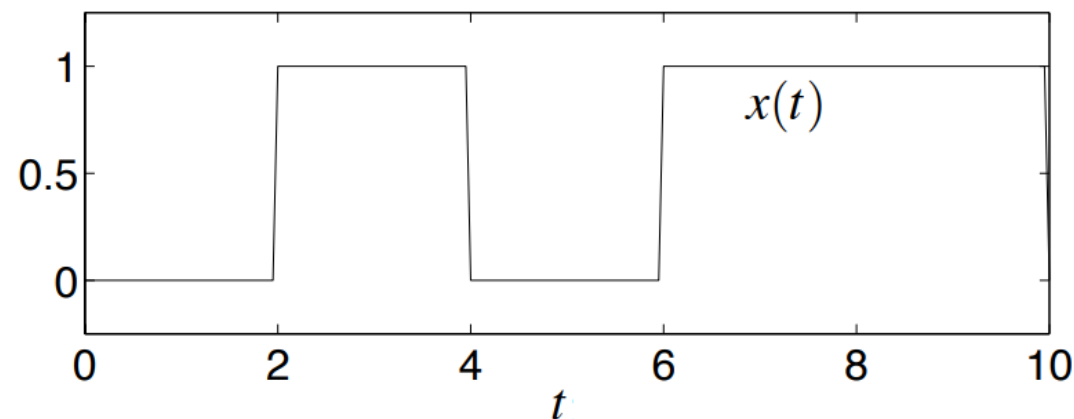


13.7 Convolution integral (f)

- Convolution is important in real-word applications:

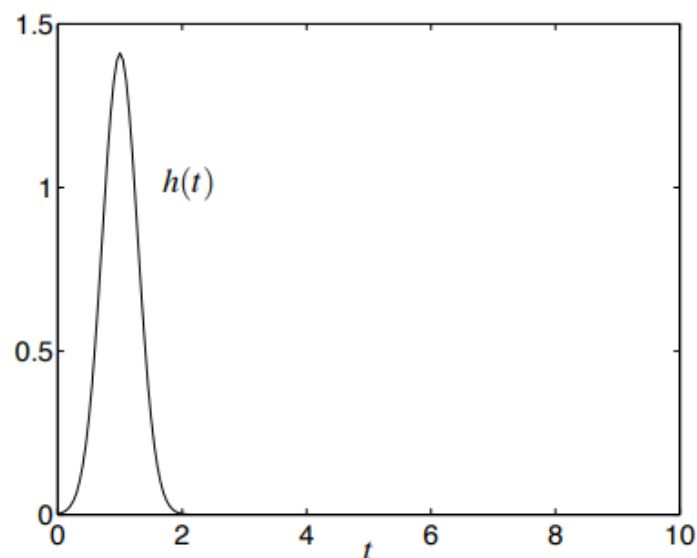


Simple signaling at 0.5 bit/sec; Boolean signal 0, 1, 0, 1, 1, ...

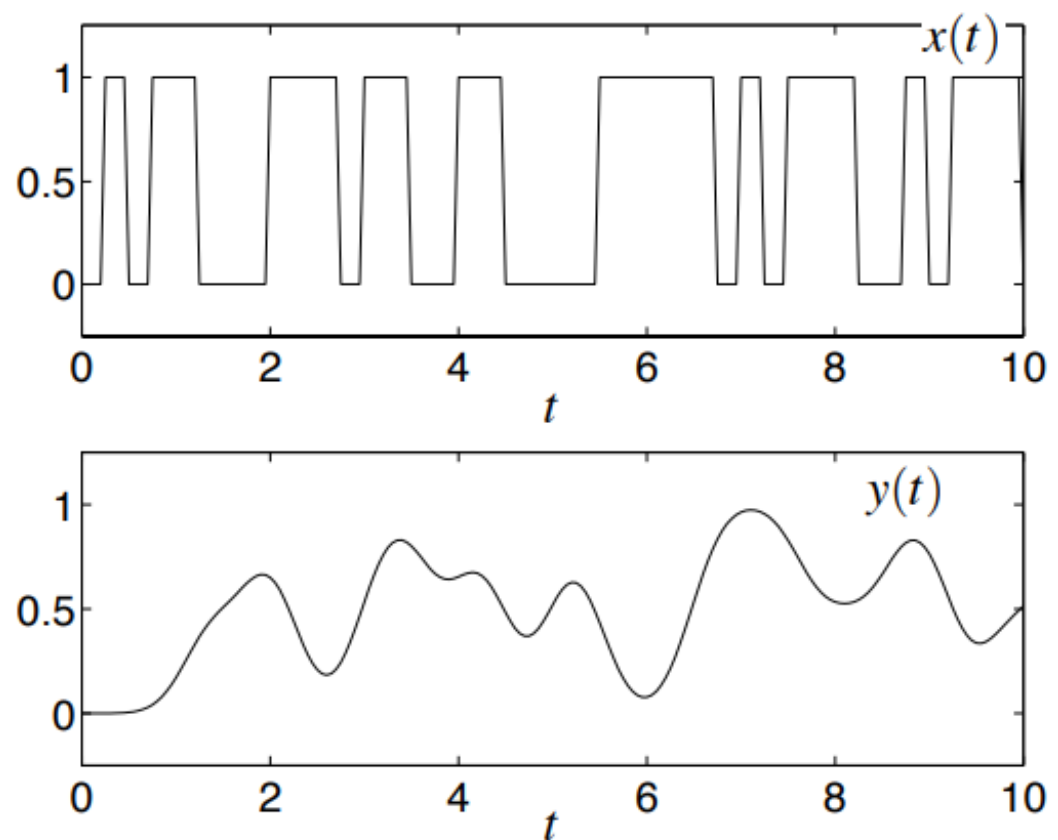


13.7 Convolution integral (g)

- Convolution is important in real-word applications:



Simple signalling at 4 bit/sec; same Boolean signal



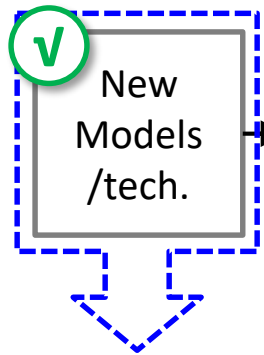
Lesson 13 Introduction to the Laplace transform

13.1-13.2	Introduction	Slides 06-13
13.3	Laplace transform: what is it?	Slides 15-26
13.4-13.5	Properties of Laplace transform	Slides 28-42
13.6	Inverse Laplace transform	Slides 44-67
13.7	*Convolution Integral	Slides 69-75
13.8	Summary	Slides 77-78

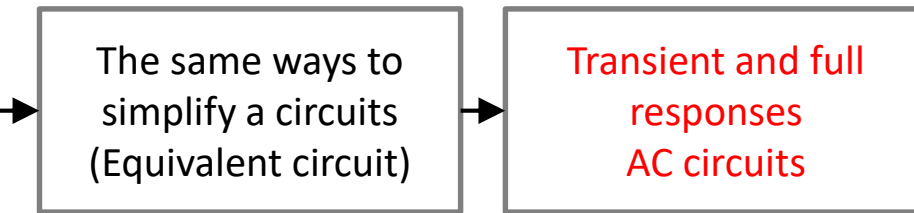
13.8 Summary and assignment (a)

- EE104 road map

Lesson 13



Lesson 14



1. Laplace transform
2. Inverse Laplace transform

TABLE 15.2

Laplace transform pairs.*

$f(t)$	$F(s)$
$\delta(t)$	1
$u(t)$	$\frac{1}{s}$
e^{-at}	$\frac{1}{s + a}$
t	$\frac{1}{s^2}$
t^n	$\frac{n!}{s^{n+1}}$

TABLE 15.2

Laplace transform pairs.*

$f(t)$	$F(s)$
te^{-at}	$\frac{1}{(s + a)^2}$
$t^n e^{-at}$	$\frac{n!}{(s + a)^{n+1}}$
$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$

TABLE 15.2

Laplace transform pairs.*

$f(t)$	$F(s)$
$\sin(\omega t + \theta)$	$\frac{s \sin \theta + \omega \cos \theta}{s^2 + \omega^2}$
$\cos(\omega t + \theta)$	$\frac{s \cos \theta - \omega \sin \theta}{s^2 + \omega^2}$
$e^{-at} \sin \omega t$	$\frac{\omega}{(s + a)^2 + \omega^2}$
$e^{-at} \cos \omega t$	$\frac{s + a}{(s + a)^2 + \omega^2}$

13.8 Summary and assignment (b)

Problems

15.9, 15.25, 15.31, 15.34, 15.39

- (a) A certain amount of practicing is necessary to sharpen your mind and to deepen your understanding.
- (b) There are plenty of problems in the textbook, try to complete as many as you can.